

Facilities Division	Effective Date: February 1, 2013	Document Number: SLAC-I-708-406-001-00	Revision: R002
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1.0 Purpose

Construction Project Requirements for Preparing Tree Protection Plan

SLAC National Accelerator Laboratory is operated by Stanford University for the Department of Energy. Stanford University, as owner of the land, is responsible for the protection and care of the trees, shrubs, and vegetation growing or planted on the SLAC site.

Based on an agreement between the Stanford Site Office of the Department of Energy and the University Architect Office, any SLAC project that will impact or take place in close proximity to existing trees shall have a written tree protection plan. This document defines the minimum requirements for developing a project-specific plan to protect trees and other vegetation in the vicinity of any construction activity performed on the SLAC site. These requirements include parking and staging of both personal and commercial vehicles and equipment.

2.0 Definitions

Injury: Without limitation, any burning, singeing, bruising, scarring, tearing, or breaking of foliage, roots, branches, or trunk.

Root zone: The area from the trunk out to 10 feet beyond the outermost limits of the tree canopy. By comparison, the “drip line” of a tree typically extends out to the edge of the crown of the tree.

Canopy: The canopy of a tree/plant refers to the totality of its aboveground parts, including stems, branches, leaves, and reproductive structures.

SLAC-Approved Certified Arborist (SACA): a person who has obtained an arborist certification from the International Society of Arboriculture and has been approved by SLAC.

Owner: Stanford University. “Owner Representative (OR)” is a representative designated by SLAC National Accelerator Laboratory.

Facilities Construction Manager (FCM): a SLAC employee who is thoroughly familiar with and competent to provide direct construction oversight of the type of work involved, and who shall oversee all construction project work to ensure compliance with this guideline and the Project as a whole.

Facilities Service Manager (FSM): (previously a UTR); A SLAC employee who is thoroughly familiar with and competent to oversee the performance of the Arborist, pruning, protection, tree and shrub removal and/or tree and shrub replacement.

International Society of Arboriculture (ISA): is the organization that sets arboricultural guidelines, which often influence the ANSI regulations. This organization is also responsible for the training, education and certification of arborists.

Tree and Shrub Protection Guideline

Facilities Division	Effective Date: February 1, 2013	Document Number: SLAC-I-708-406-001-00	Revision: R002
----------------------------	--	--	--------------------------

Diameter [at] Standard Height (DSH): the measurement of the diameter of a tree trunk measured at standard height (4.5 feet above the ground).

Heritage Tree: A tree or group of trees of historical significance. In particular, a redwood tree (*Sequoia sempervirens*), a sequoia (*Sequoiadendron giganteum*) or an oak tree (genus *Quercus*) which is native to California and has a trunk with a circumference of 31.4 inches (diameter of 10 inches) or more, measured 48 inches above natural grade. Any tree other than oaks, redwoods, and eucalyptus which has a trunk with a circumference of 47 inches (diameter of 15 inches) or more, measured 48 inches above natural grade. Trees with more than one trunk shall be measured at the point where the trunks divide.

Memorial Tree: Any tree that has been planted in memory of a SLAC employee and is maintained by SLAC Facilities.

3.0 Minimum Requirements for Tree Protection / Removal Plan

- Design Drawings Showing Project Boundaries
- Sub-contractor access routes, material and equipment laydown area and parking
- Utility routing (location of existing and future utilities)
- Excavations (limits of grading and excavations)
- Existing trees, overhead view of drip lines
- Updated tree inventory and location map
- Pre-work photographs of construction site(s)
- SLAC Arborist Tree/Shrub evaluation report on project impact to trees & shrubs

All existing trees, shrubs and groundcovers to remain shall be indicated on the drawings. Removal of trees to be relocated shall be scheduled for completion before construction begins. Trees to remain during construction shall be protected in place to the extent necessary (see Section 5.0: Tree Protection and Removal plan).

Once the above information is gathered, it should be reflected on the project plans including work procedures to mitigate tree protection. This will provide a SLAC Approved Certified Arborist (SACA) and BIO plan reviewers with needed information for the required approvals (SLAC and SU), prior to initiating any work on or associated with the project site.

3.1 Related Work

The following types of work typically involve tree and shrub protection:

- Basic Mechanical Materials and Methods
- Earthwork
- Irrigation System
- Landscape Design

Facilities Division	Effective Date: February 1, 2013	Document Number: SLAC-I-708-406-001-00	Revision: R002
--------------------------------	--	--	--------------------------

- Planting
- Site Preparation
- Site Restoration and Rehabilitation

4.0 Quality Assurance

4.1 General Responsibility

The Sub-Contractor shall be directly responsible for protection and welfare of all existing trees, shrubs and groundcovers within and around the project area that are to be saved, in addition to new plantings and any relocated trees. This responsibility shall continue throughout the full construction period until the entire project is completed and accepted by the Owner or Owner Representative, and through completion of the warranty period of one year.

Sub-Contractor shall not take any action that may result in injury or death of any tree or other vegetation, including but not limited to excessive pruning, root-zone soil compaction, cutting, girdling, poisoning, over-watering, unauthorized relocation or transportation, trenching, excavating, altering the grade, or paving within the root zone.

Exceptions deemed necessary shall be done under the guidance and review of a SACA to minimize negative impacts to the health of the tree.

4.2 Qualifications of Tree Workmen

If tree trimming is recommended by the SACA: A qualified Tree Service Company, supervised by a Facilities Service Manager (FSM) or designate (who is thoroughly familiar with the type of work involved, and shall direct all protection and trimming work), will oversee this service for compliance to the SACA's recommendations. No more than 1/4 of the foliage of a tree shall be removed in a 12-month period, except in the case of impending danger or hazardous condition.

4.3 Job Conditions and Construction Requirements

Prior to performing any work that may impact trees or shrubs, the Sub-Contractor or Project Manager shall call the Facilities FSM to request a site meeting with a SACA. This meeting shall occur prior to construction, demolition or access of any nature on the project site and related areas, including access routes, laydown & staging areas.

This meeting shall establish the conditions and methods for preserving all existing trees and identifying the plant materials to be saved.

4.4 Sequencing Schedule

Tree and Shrub Protection Guideline

Facilities Division	Effective Date: February 1, 2013	Document Number: SLAC-I-708-406-001-00	Revision: R002
----------------------------	--	--	--------------------------

Coordinate and cooperate with other trades to enable the work to proceed as rapidly and efficiently as possible. Compile the information into a single schedule for reference. Protective fencing shall be in place and approved by the SACA or FSM before any work begins at the site, including material delivery and storage.

5.0 Submittals

5.1 Tree Protection / Removal Plan

The following are requirements for a tree protection and Removal plan:

- Contact SLAC Facilities Division for identification of all significant trees and shrubs in and around the project area.
- Develop a project-specific plan that shows an accurate representation of all existing trees over 3" in Diameter at Standard Height (DSH), any SLAC-designated young specimen trees and significant trees and/or shrubs of any diameter, with clear indications of which are to be removed, relocated or retained on the site. This plan shall include the botanical name, DSH and canopy diameter for every plant specimen.
- Laydown, staging and parking areas must be shown on the plans and approved by a SACA if any of the above could impact root systems or canopies. Tree protection requirements apply to all trees in laydown, staging and parking areas, as well as to trees within the project limits.
- This plan shall include quantity and frequency of watering to maintain the existing trees and shrubs as well as new (planted or unplanted) trees and shrubs in the construction area. This plan must also be reviewed and approved by a SACA.
- Remaining trees at the project site must have protective fencing installed per the tree protection drawing included in the plan set.
- The protective fencing plan must indicate the outer edge of the root zone (see 2.0 Definitions, above) for each tree in the project area. If project constraints do not allow for fencing at the outer edge of the root zone, fencing must be placed as close to this as possible and approved by a SACA.
- Protective fencing shall be chain-link or orange plastic barricade fencing (on secure footings or imbedded), or as required by the SACA.
- This plan shall be submitted with the project plans for review and must be approved by a SACA, SLAC and SU before construction or pre-construction activity begins, including storage and/or delivery of equipment, temporary construction offices and/or materials, demolition, etc. This Plan shall be included in a field set of drawings and kept onsite at all times for reference during construction.

5.2 Other Submittals

- Watering schedule, where interruption of irrigation systems will exceed one watering period, as approved by the SACA.

Tree and Shrub Protection Guideline

Facilities Division	Effective Date: February 1, 2013	Document Number: SLAC-I-708-406-001-00	Revision: R002
----------------------------	--	--	--------------------------

- Tree Removal Plan to include the tree identification number, type of tree, size, and whether it is a Heritage Tree, Memorial Tree, or other special status.
- A Replanting Plan listing each tree removed (from above demolition plan), species of new tree to be planted, size and approved SLAC new installation location.

6.0 Protection

- Laydown, staging and parking areas must be shown on the plans and approved by a SACA if any of the above will impact tree or shrub root or canopy protection zones. All tree protection guidelines apply to trees in laydown, staging and parking areas as well as to trees within the project limits.
- Construction materials, equipment, and personal vehicles shall not be stored, parked or temporarily placed in the root zone of any tree. Nothing shall be stored or placed temporarily within protective fencing, to avoid soil compaction, soil contamination, or physical damage to trees. Root zones of trees shall not be driven over. Provide alternative routes for construction traffic of any kind, including cars, people, tractors, equipment, cranes, or any other traffic and all staging or storage areas.
- During the course of construction the Sub-Contractor shall take all necessary precautions, as outlined herein, to protect from stress, injury or death to all existing trees, shrubs and groundcovers to be preserved or relocated. Protection shall be given to the roots, trunk, and foliage of all existing plant materials to remain.
- Trees that have been injured must be repaired immediately with the approval of the SACA. All costs associated with mitigation of damage to trees are to be paid by the sub-contractor.
- Protect overhanging tree canopies from construction damage. If drive aisles are anticipated under low canopies, call for an evaluation by a SACA to determine appropriate measures.
- Existing trees shall be monitored weekly and irrigated by the contractor as needed during the course of construction.
- No lime (calcium oxide) or other soil treatment shall be applied without the prior approval of a SACA.
- No rinsing or cleaning of equipment or dumping of construction-related liquid materials shall be allowed in the tree root zone, or in an area that drains into the root zone. Care shall be taken when cleaning equipment. There shall be no storage of dumpsters or accumulated debris from demolition on or around the root zones of existing trees and shrubs.
- All trenching shall conform to the following guidelines:
- The Facilities Construction Manager (FCM) is required to be present to supervise any trenching, digging or excavation of any kind within the root zone of any tree to make sure that the sub-contractor is following the excavation and Tree protection plans.
- Roots larger than 2 inches in diameter shall not be severed without a SACA approval. Tunneling or boring under roots is preferred over cutting.

Tree and Shrub Protection Guideline

Facilities Division	Effective Date: February 1, 2013	Document Number: SLAC-I-708-406-001-00	Revision: R002
----------------------------	--	--	--------------------------

- If root cutting is the only approved option, then cut shall be pruned cleanly, not torn.
- Mechanical digging within a tree's root zone shall be avoided. If necessary, hand digging shall be used for any trenching within the tree's root zone unless otherwise approved by a SACA. The preceding guidelines shall be considered minimum requirements. The greater the distance of tree protection provided, the greater the instance of tree success in construction areas.

6.1 Tree Protection Fencing

- All trees retained in place or to be relocated shall be protected with semi-permanent chain-link fences or orange plastic barricade fencing (on secure footings or imbedded) as directed by the SACA/FSM. The fences shall be placed at the outer edges of the root zone.
- The SACA/FSM must approve fencing after it is in place and before any construction activity begins, including delivery of materials or temporary construction offices and site preparation activities.
- Fences shall be erected before construction begins and remain in place until final inspection of project, unless approval by the SACA/FSM is given for temporary relocation to complete work approved in the plans.
- If the trees are in a small area and cannot be fenced as planned, the tree trunk must be wrapped with orange plastic barricade fencing, or other protection as agreed upon with the SACA. Caution must be used so as not to damage the trunk or any branches. Major scaffold limbs may require the same treatment as the above, or as directed by the SACA.
- During the course of construction, if relocation of the fence is required to facilitate construction, the Sub-Contractor shall do so only with the approval of a SACA before any work begins, at no additional expense to the Owner.
- Approval by the SACA for work within the fenced area shall not release the Sub-Contractor from any of the provisions specified herein for the protection of existing trees to be preserved. SACA approval can be in writing or by email. No verbal approvals are authorized.

6.2 Excavation

Excavation around trees and shrubs:

- Excavation within the root zone of trees shall be done only where absolutely necessary and only with a SACA approval and direct supervision by the FCM.
- Digging, tunneling or boring under or around roots shall be used instead of root pruning wherever possible, as agreed upon by the SACA. Where trenching for utilities is required within root zone, tunneling under and around roots shall be by hand digging. Roots larger than 2" in diameter shall not be cut without prior approval from a SACA.
- Where excavation for new construction is specifically required in approved plans within root zones of trees, hand excavation, or mechanical boring that goes only

Tree and Shrub Protection Guideline

Facilities Division	Effective Date: February 1, 2013	Document Number: SLAC-I-708-406-001-00	Revision: R002
--------------------------------	--	--	--------------------------

under roots without breaking the soil surface, shall be employed with the approval of a SACA to minimize damage to root systems.

- Existing grade of tree at its canopy shall be marked with a water-based non-toxic paint before construction begins.
- There shall be no grade changes within a minimum of 10 feet of the trunk of existing trees, and preferably none within the entire canopy, except as approved by the SACA.
- If roots larger than two inches in diameter are encountered, they shall be exposed beyond excavation limits as required for pruning. Roots shall be hand pruned under direction and approval of a SACA and treated as exposed roots as stated below in item 8.
- All roots that need to be cut shall be pruned cleanly, not torn, under direction and approval of a SACA. All damaged roots over two inches in diameter shall be pruned leaving a clean face and sealed with a non-oil based, alcohol-based, wood sealer that disinfects the roots. Sealer must be pre-approved by the SACA and SLAC's Building Inspection Office (BIO).
- Exposed, cut or broken roots shall not be allowed to dry out before permanent backfill is placed, as directed by SACA. Temporary earth cover shall be provided, or roots shall be packed with wet peat moss or four layers of wet, untreated burlap and temporarily supported and protected from damage until permanently relocated and covered with backfill. The cover over the roots shall be wetted to the point of runoff daily.

6.3 Prohibited Activities

The following activities are prohibited within the root zone of any existing tree to be preserved, or on existing or new lawn areas and groundcover areas:

- Storage, driving or parking of automobiles or other vehicles or equipment. Root zones of trees shall not be driven over. Provide alternative routes for construction traffic of any kind, including cars, people, tractors, equipment, cranes, or any other traffic and all staging or storage areas. Any exceptions must be approved by a SACA.
- Stockpiling or temporary storage of building materials, soils, dumpsters, accumulated debris, refuse or excavated materials are prohibited in the root zone of a tree or bush to remain.
- Skinning or bruising of bark.
- Use of trees as support posts, power poles, or signposts; anchorage for ropes, guy wires, or power lines; or other similar functions.
- Dumping of materials on or around trees and roots. Such materials include but are not limited to paint, lime, paint thinner, petroleum products, dirty water, or other deleterious materials.

Tree and Shrub Protection Guideline

Facilities Division	Effective Date: February 1, 2013	Document Number: SLAC-I-708-406-001-00	Revision: R002
----------------------------	--	--	--------------------------

- Damage to trunk, limbs, or foliage caused by maneuvering vehicles or stacking material or equipment too close to the tree.
- Compaction of root zones by excessive foot traffic, movement of trucks or grading machines; storage of equipment, gravel, earth fill, or construction supplies; etc.
- Excessive water or heat from equipment, utility line construction.
- Damage to root systems from flooding, erosion, puddling or continuous running water, and excessive wetting and drying resulting from dewatering and other operations.

7.0 Tree Removal

- Dead and damaged trees that are determined by a SACA to be incapable of restoration to normal growth pattern or lifespan shall be removed at the expense of the sub-contractor.
- Trees designated for removal shall be removed to a point at least 1 foot below the lowest level of subgrade upon which fill will be placed.
- Soil within a radius of 10 feet of the removed tree under paved area shall receive treatment as recommended by a SACA before further preparation of the subgrade.
- Trees designated as impacting the construction zone that cannot be protected or fenced off will need to be relocated or replaced.

8.0 Tree Replacement

Tree Replacement will be included in the project and performed by the subcontractor.

Tree replacement will depend on the type of the existing tree (heritage, memorial, other). Following the type designation, the following standards will apply.

- Memorial Tree: One-for-one replacement; the type will be like-for-like.
- Heritage Tree: three-for-one replacement; tree type will be based on SACA recommendation with SLAC and Stanford University approval.
- Other trees will be one-for-one replacement; tree type will be based on SACA recommendation with SLAC and Stanford University approval.
- Eucalyptus: will be replaced one-to-one with native species approved by SLAC.
- Replacement tree locations will be approved by SLAC before any planting can occur.

Sub-contractor will be responsible for maintaining the new planting for one year after installation at the subcontractors' expense.

9.0 Pruning of Trees

Tree and Shrub Protection Guideline

Facilities Division	Effective Date: February 1, 2013	Document Number: SLAC-I-708-406-001-00	Revision: R002
----------------------------	--	--	--------------------------

- In concurrence with the SACA, ascertain the limbs and roots to be trimmed and clearly mark them to designate the approved point of pruning.
- A SACA shall be engaged to approve the removal of branches from trees and large bushes which are to remain, if required to clear for new construction.
- Prune limbs evenly, using proper tools and skilled work persons under supervision of a Facilities Services Manager, to minimize damage to the tree.
- In the case of root cuts, apply wet burlap or other protection, approved by the SACA, to prevent drying out, and maintain in a wet condition as long as necessary for temporary protection as determined by the SACA.

10.0 Watering

- Sub-Contractor will conform to the approved watering plan for remaining and new planted or unplanted trees and shrubs in construction zone.
- Sub-contractor will provide ample water supply of potable quality and sufficient quantity for all operations required by the plan.
- During construction the existing site surface drainage patterns shall not be altered within the area of the root zone of any tree.
- Sub-Contractor shall not alter the existing water table within the root zone.
- Sub-Contractor shall continuously observe existing vegetation onsite and complete necessary measures to maintain healthy living conditions for existing trees, shrubs and groundcover to be preserved. Such measures include promptly notifying the SACA through the FSM if any decline is observed, so mitigation measures can be determined. These shall include, but not be limited to, periodic washing of leaves for the removal of dust, soil aeration and supplemental or interim irrigation.
- Where traffic in a tree's root zone is necessary, the root zone shall be tested by the contractor for compaction and corrective measures implemented or mitigated as necessary and approved by a SACA, before any construction activities commence. Depending on the soil conditions, two to three inches of mulch, wooden planks, steel plates, or other protective material shall be laid down as specified by a SACA to reduce compaction.

11.0 Repair Compensation

Damage to existing tree canopies or roots over 2 inches in diameter shall be immediately reported to the Project Manager, the SACA and repaired immediately at the Sub-Contractor's expense,

Repairs shall be made promptly (per SACA recommendations) after damage occurs to prevent progressive deterioration of damaged trees. The FSM shall supervise repair of trees damaged by construction operations.

Any tree slated to remain which is inadvertently damaged beyond repair or destroyed, as determined by a SACA, by the the Sub-Contractor's negligence or failure to provide adequate protection, shall be compensated for at the Sub-Contractor's expense and paid to SLAC Value will be dependent on its diameter, species, location, condition and other factors in accordance

Facilities Division	Effective Date: February 1, 2013	Document Number: SLAC-I-708-406-001-00	Revision: R002
----------------------------	--	--	--------------------------

with the schedule of values in ISA & Council of Trees and Landscape Appraisers most current printing of *Guide for Plant Appraisal*.

The following are three examples of the maximum costs that will be expected for a coast live oak (*Quercus agrifolia*) with a 6", 18" and 24" DSH (trunk diameter at standard height). The costs were calculated by using the Replacement Cost Method Calculation from the *Guide for Plant Appraisal, 9th Edition* published by ISA & Council of Trees and Landscape Appraisers. Costs rise as time passes and this estimate was calculated in fall of 2008.

6"	\$ 2,350
12"	\$21,300
24"	\$26,800

A sub-contractor may be assessed fees of this magnitude if they severely damaged and/or killed a tree as determined by the SACA. This cost does not include maintenance such as monitoring and watering necessary to re-establish the new tree, which would also be the responsibility of the sub-contractor for up to 90 days begin after a substantial completion walk-through.

Damaged tree limbs or trees which have died as a result of injury during construction shall remain the property of Stanford University and shall remain or be removed at the expense of the Sub-Contractor as directed by the SACA.

12.0 Clean-Up

At close of construction in each area, all protective barriers and any accumulated debris shall be carefully removed without damaging trees or other vegetation at the direction of the SACA. All barrier materials shall be transported offsite at no additional expense to Owner.

All grades and areas of soil compaction shall be repaired, and all damaged plant materials shall be restored as determined by and in conjunction with the SACA and the FSM.

13.0 Warranty

Sub-Contractor shall warrant that all trees, shrubs and groundcover covered by the provisions of this Section will be healthy and in flourishing condition of active growth 1 year from the date of Substantial Completion. Where there has been evidence of neglect or violation of tree protection, the warranty shall extend for an additional year.

During the warranty period the Sub-Contractor shall be liable for damages to all plant material including trees, shrubs and groundcover. Sub-Contractor shall pay compensation or replace plant material if a tree or significant shrub to remain is destroyed or damaged, so that in the judgment of the SACA it is not healthy and thriving or not able to recover. The replacement plant materials shall closely match in size, color, and variety, the damaged plantings. If replacement of trees is not desired by the SACA, damages will be assessed at the value of the tree as determined by ISA's most current printing of the *Guide for Plant Appraisal* (listed in 1.05, part A of this document).

Tree and Shrub Protection Guideline

Facilities Division	Effective Date: February 1, 2013	Document Number: SLAC-I-708-406-001-00	Revision: R002
----------------------------	--	--	--------------------------

Damaged shrubs and groundcover shall be replaced in kind and shall match sizes removed or be otherwise agreed upon by the SACA.

Sub-Contractor will not be held responsible for failures due to neglect by the Owner, vandalism, etc., during the warranty period only if such conditions are reported immediately in writing to and reviewed onsite by the SACA and the damage is documented and photographed.

14.0 Reference Standards

Published specifications, standards, tests, or recommended methods of trade, industry, or governmental organizations apply to the work covered in this document.

International Society of Arboriculture (ISA) *Guide for Plant Appraisal*, 9th edition, prepared by the Council of Tree and Landscape Appraisers (CTLA), ISBN # 1-881956-25-3.

International Society of Arboriculture (ISA) Best Management Practices Series. These are booklets bundled with ANSI 300 Guidelines:

- *ANSI A300 - Best Management Practices Fertilization Combo*
- *ANSI A300 - Best Management Practices Integrated Vegetation Management Combo*
- *ANSI A300 - Best Management Practices Planting Combo*
- *ANSI A300 - Best Management Practices Pruning Combo*
- *ANSI A300 - Best Management Practices Support Systems Combo*

15.0 Plans Cited in this Document

- Tree Protection Plan (primary document)
- Tree Removal Plan (separate or apart of the Protection Plan)
- Tree Protective Fencing Plan
- Tree Re-Planting Plan
- Sequencing Schedule
- Watering Schedule

16.0 Revision History

Rev No.	Effective Date:	DCO No.
R002	2/01/2013	1002
R001	10/26/2010	N/A